The Midnight Library Mystery

Whodunnit? · Foundations of Algebra · Teacher Guide and Answer Key

Unit	Time	Supplies	Group size
Unit 1: Foundations of Algebra	20–40 minutes	Pencil, scrap paper, scissors	Teams of 2–3 (or solo)

Skills practiced: order of operations, evaluating expressions, combining like terms, the distributive property, and writing expressions from words.

General procedures

1. Give each student the **student sheet** (story and case board) and the **Show Your Work** page.
2. Print the **Clue Cards** (pages 3–7 of the student packet) and cut them apart on the dashed lines. Post them around the room.
3. Students visit the cards in any order. At each card they solve the problem, find their answer among the four choices, read the clue beside it, and cross things off the case board. **Every wrong choice leads to a false clue** that clears the real culprit, so a mistake shows up when the board does not narrow down to one suspect, one place and one method.
4. When a team thinks it knows, they bring you the three names. If they are wrong, let them hunt for the card where something went wrong.

Solution: Professor Hale took it, in the Map Room, using this method: Rolled up in a poster tube.

The reveal (read aloud): When the lights flickered, **Professor Hale** slipped into the **Map Room**, rolled the atlas into a **poster tube** labeled “Book Tour Posters,” and strolled toward the exit. Every alibi and every method you cleared pointed straight at him. Brilliant detective work!

Clue map: what each correct clue clears

Card	Clears
1	Suspect: Mr. Patel · Place: Front Desk
2	Method: Hidden in a book cart
3	Suspect: Ms. Greene · Place: Reading Room
4	Method: Tucked in a backpack
5	Suspect: Officer Ruiz · Place: Rare Books Vault
6	Method: Slipped into a board-game box
7	Suspect: Jada Brooks · Place: Children’s Corner
8	Method: Swapped with a fake cover
9	Suspect: Mr. Fox · Place: Computer Lab
10	Method: Lowered out a window in a basket

Answer Key

The Midnight Library Mystery · Correct choice and worked solution for each card

#	Problem	Correct choice	Worked solution	Clue it leads to
1	Evaluate: $18 - 2(3 + 4)$	**C.** 4	$2(7) = 14$ $18 - 14 = 4$	Mr. Patel was checking out books at the Front Desk the whole time. Twelve families saw him there.
2	Evaluate: $3^2 + 4 \cdot 5 - 6$	**A.** 23	$9 + 20 - 6$ $= 23$	The book cart was searched at the door. It held only picture books.
3	Evaluate when x = 5: $4x - 7$	**B.** 13	$4(5) - 7$ $20 - 7 = 13$	Ms. Greene was leading a book club in the Reading Room. Her group never let her leave.
4	Evaluate when a = −3 and b = 4: $a^2 - 2b$	**D.** 1	$(-3)^2 - 2(4)$ $9 - 8 = 1$	Every backpack was checked at the exit. No atlas would even fit.
5	Simplify: $5x + 3 - 2x + 8$	**D.** $3x + 11$	$(5x - 2x) +$ $(3 + 8)$ $3x + 11$	Officer Ruiz was locked inside the Rare Books Vault doing inventory. The vault log shows the door never opened.
6	Simplify: $4(2x - 3)$	**C.** $8x - 12$	$4(2x) - 4(3)$ $8x - 12$	The board-game boxes were all opened for Game Night. Nothing but game pieces inside.
7	Simplify: $3(x + 4) - 5x$	**D.** $-2x + 12$	$3x + 12 - 5x$ $-2x + 12$	Jada Brooks was reading to toddlers in the Children's Corner. A dozen parents took videos.
8	Write an expression: 5 less than 3 times a number x	**C.** $3x - 5$	3 times a number: $3x$ 5 less than that: $3x - 5$	The fake cover in the case is a decoy the library always uses. The real atlas was never swapped.
9	Evaluate: $\lvert -8 \rvert + (-3)(4)$	**D.** -4	$8 + (-12)$ $= -4$	Mr. Fox was running a trivia contest from the Computer Lab. The livestream never shows him leaving.
10	Simplify: $-2(3x - 5) + 7x$	**B.** $x + 10$	$-6x + 10 + 7x$ $x + 10$	All the windows are painted shut for the winter. Nothing went out in a basket.

Wrong Choices

The Midnight Library Mystery · The likely mistake behind each wrong choice, and the false clue it leads to

Card	Choice	Likely mistake	False clue
1	A. 112	subtracted before multiplying	Professor Hale was signing books at the Front Desk the whole time.
1	B. 16	multiplied only the 3 by 2	The poster tubes were checked. Only posters inside.
1	D. 23	ignored the multiplication	Mr. Patel took a long break in the Map Room.
2	B. 20	used $3^2 = 6$	Professor Hale never left the Reading Room.
2	C. 59	worked left to right, adding before multiplying	Jada Brooks was seen hurrying out of the Map Room.
2	D. 5	subtracted $5 - 6$ first	Officer Ruiz was patrolling the Computer Lab.
3	A. 38	read 4x as 45	A guard checked every poster tube at the door.
3	C. -8	subtracted before multiplying	Ms. Greene was alone in the Map Room for ten minutes.
3	D. 27	added 7 instead of subtracting	The Map Room was locked all evening.
4	A. -17	used $-3^2 = -9$	Mr. Fox carried a large backpack out to his car.
4	B. -14	used $a^2 = 2a$	Professor Hale spent the night in the Children’s Corner.
4	C. 17	added instead of subtracting	Someone saw a book cart rolling toward the loading dock.
5	A. $14x$	combined unlike terms	Officer Ruiz left the vault to check the Map Room.
5	B. $7x + 11$	added 2x instead of subtracting	Professor Hale was in the Rare Books Vault with the guard.
5	C. $3x + 5$	subtracted the 8	The window basket was found sitting on the lawn.
6	A. $8x - 3$	distributed to the first term only	Jada Brooks had a poster tube under her arm.
6	B. $6x - 7$	added 4 instead of multiplying	Professor Hale was helping in the Computer Lab.
6	D. $8x + 12$	sign error	A board-game box felt much too heavy.
7	A. $8x + 12$	added 3x and 5x	Mr. Patel walked out with a very large backpack.
7	B. $-2x + 4$	did not multiply the 4 by 3	Professor Hale was reading in the Children’s Corner.
7	C. $2x + 12$	sign error	The fake cover had been swapped with the real one.
8	A. $5 - 3x$	subtracted in the wrong order	Mr. Fox was spotted near the Map Room door.
8	B. $3(x - 5)$	took 5 from x first	Ms. Greene asked to borrow a poster tube.
8	D. $5x - 3$	switched the numbers	Professor Hale never touched a poster tube.
9	A. -20	used $\lvert -8 \rvert = -8$	Officer Ruiz left the vault for an hour.
9	B. 20	used $(-3)(4) = 12$	The poster tube was empty when security looked.
9	C. 9	added instead of multiplying	Jada Brooks left the Children’s Corner early.
10	A. $x - 10$	sign error: $-2 \times -5 = +10$	A basket was lowered from the Reading Room window.
10	C. $13x + 10$	sign error on 6x	Professor Hale was at the Front Desk with Mr. Patel.
10	D. $x + 5$	did not multiply the 5 by -2	Mr. Fox left the livestream for twenty minutes.